Exact regularization thresholds for integer quadratic transport

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Abstract

We study optimal transport with integer flows and heterogeneous quadratic regularization. Our main result gives the exact regularization threshold up to which a unique linear-optimal plan remains integer-optimal. A residual-cycle formula separates the continuous directional derivative from a positive finite-step correction, so the integer threshold strictly exceeds the continuous threshold whenever the latter is finite. Under population replication, normalized integer thresholds decrease to the continuous threshold; we identify the first-order correction and an exact formula valid beyond an explicit replication index. We also establish monotone comparative statics for integer optima: increasing a connection's own linear or quadratic cost weakly reduces its flow, and increasing regularization weakly reduces total quadratic congestion. A discrete convex profile gives exact cost thresholds and unit adjustments, while a counterexample rules out monotonicity of individual flows in the regularization parameter. Finally, feasible rounding yields sharp second-order value and first-order distance bounds.

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1 Introduction

Integer margins guarantee integral vertices in linear optimal transport, but quadratic regularization can select a fractional plan. When flows represent indivisible units, the relevant problem minimizes the regularized objective over integer transport plans. We ask when a linear-optimal plan first loses integer optimality, how that threshold changes under population replication, and which comparative statics survive indivisibility.

Our main contribution is an exact persistence threshold for a unique linear-optimal plan. Theorem 2 expresses the integer threshold through feasible residual cycles and isolates the finite-step correction absent from the continuous problem. This correction strictly delays the first transition and can change its determining cycle. Theorem 5 gives a monotone replication law, an explicit index beyond which one formula is exact, and the first-order threshold error. Theorem 3 specializes the cycle criterion to pure group assignments with arbitrary positive quadratic weights.

The integer comparative statics are stated without assuming a unique regularized optimum. Increasing a connection's own linear or quadratic coefficient weakly reduces its flow, while increasing regularization weakly lowers total quadratic congestion and weakly raises linear transport cost (Proposition 6). A discrete convex profile further gives exact cost thresholds and adjustments of one unit (Proposition 7). Individual flows can reverse direction as regularization increases. Finally, Proposition 8 gives sharp second-order value and first-order distance bounds under grid refinement. Accurate objective values can therefore coexist with larger errors in the regularization level at which the allocation changes.

The optimization machinery is classical. Separable convex-cost flows admit marginal-cost algorithms and residual-cycle certificates (Hochbaum & Shanthikumar, 1990; Minoux, 1986; Murota, 2016); discrete convexity also describes sensitivity to costs and capacities (Murota & Shioura, 2005). We use these established criteria to derive the finite-step threshold correction and its replication law. The rounding and monotonicity results are supporting consequences of integrality, convexity, and optimality; the quantitative coordinate thresholds use a cycle exchange argument.

For continuous problems, González-Sanz and Nutz (2025, Theorem 2.5) characterize the exact threshold of quadratic regularization over an arbitrary polytope. Their result allows any inner product and therefore includes heterogeneous quadratic weights. The continuous formula below is its transport-cycle specialization. The additional term in the integer formula determines a different critical cycle in some instances and yields the replication correction studied here. Feasible rounding uses the marginal and degree preservation underlying dependent rounding (Gandhi et al., 2006); no new rounding algorithm is needed.

The model of Gallardo et al. (2025) penalizes deviations from prescribed margins. Here the margins hold exactly, and the analysis compares continuous and indivisible flows for the same objective. Section 2 states the required flow criterion. Sections 3 and 4 give the threshold formulas and replication law. Section 5 develops integer comparative statics, and Section 6 gives approximation bounds. Appendix A records existence, the continuous solution and its derivatives, and the general two-by-two formula.

2 Integer transport and the flow criterion

Let $\mu \in \mathbb{Z}^n_{++}$ and $\nu \in \mathbb{Z}^m_{++}$, with $n, m \geq 2$ and $\sum_i \mu_i = \sum_j \nu_j$. Write

$$P = \left\{ x \in \mathbb{R}^{n\times m}_+ : \sum_j x_{ij} = \mu_i, \quad \sum_i x_{ij} = \nu_j \right\}. \tag{1}$$

For $q \in \mathbb{Z}_{++}$, set $P_q = P \cap q^{-1}\mathbb{Z}^{n\times m}$. Our primary feasible set is $P_1 = P \cap \mathbb{Z}^{n\times m}$. More generally, $qx \in qP_q$ transports integer populations $q\mu, q\nu$; q is a replication factor and $1/q$ the normalized grid size. The sets P_q need not be nested. Index connections by $e = (i, j)$ and consider

$$F_t(x) = c \cdot x + tQ(x), \qquad Q(x) = \sum_e b_e x_e^2, \qquad b_e > 0, \quad t \geq 0. \tag{2}$$

We minimize F_t over P_q and compare with its relaxation over P. The continuous optimizer $x^{\mathbb{R}}(t)$ is unique for $t > 0$. Grid optima exist because P_q is nonempty and finite; strict convexity does not imply their uniqueness.

For $x \in P_q$, a feasible signed simple cycle g alternates between origin and destination nodes. Its nonzero entries are $+1$ or -1, and $g_e = -1$ requires $x_e > 0$. Consequently, $x + g/q \in P_q$. We exclude the two-arc walk using the same connection in both directions, which makes no change in the plan.

Lemma 1 (Classical convex-flow criterion) *For $F(x) = \sum_e (c_e x_e + a_e x_e^2)$, $a_e > 0$, a plan $x \in P_q$ is optimal if and only if*

$$F(x + g/q) - F(x) \geq 0 \tag{3}$$

for every feasible signed simple cycle g. It is the unique optimum if and only if all these inequalities are strict. Any two optima differ by at most $1/q$ in each coordinate.

Proof. Use forward residual costs $c_e + a_e(2x_e + 1/q)$ and, when $x_e > 0$, reverse costs $-c_e - a_e(2x_e - 1/q)$. Their sum along g is $q[F(x + g/q) - F(x)]$. If no cycle has negative cost, shortest-path potentials give numbers u_i, v_j such that

$$\begin{aligned} c_{ij} + a_{ij}(2x_{ij} - 1/q) \leq u_i + v_j \leq c_{ij} + a_{ij}(2x_{ij} + 1/q) \qquad & (x_{ij} > 0), \\ u_i + v_j \leq c_{ij} + a_{ij}/q \qquad & (x_{ij} = 0). \end{aligned}$$

These inequalities say that x_{ij} minimizes $c_{ij}z + a_{ij}z^2 - (u_i + v_j)z$ over $z \in q^{-1}\mathbb{Z}_+$, where $\mathbb{Z}_+ = \{0, 1, \ldots\}$. Summation cancels the potential terms for every feasible competitor and proves optimality. Necessity follows from a negative cycle. Every other optimum must minimize the same one-dimensional quadratics, whose minimizers are at most two adjacent grid points. A zero-cost cycle gives a distinct optimum. Conversely, decomposing the difference of two optima into cycles conforming to its signs and using convexity shows that distinct optima require a zero-cost cycle. □

Lemma 1 is the quadratic specialization of the standard convex-flow criterion (Murota, 2016, Section 12). Increasing discrete marginal costs also give an exact minimum-cost flow formulation by expanding each connection into unit-capacity arcs. This expansion is pseudopolynomial in the total population; scaling methods avoid

explicit unit expansion under suitable cost representations (Hochbaum & Shanthikumar, 1990; Minoux, 1986).

3 Exact regularization thresholds

Assume that x^0 uniquely minimizes $c \cdot x$ over P. Total unimodularity makes x^0 an integer vertex, hence $x^0 \in P_q$ for every q. Let $\mathcal{G}$ be its finite set of feasible signed simple cycles. For $g \in \mathcal{G}$, define

$$A_g = c \cdot g > 0, \qquad D_g = -2\sum_e b_e x^0_e g_e, \qquad S_g = \sum_e b_e g_e^2 > 0. \tag{4}$$

Unique linear optimality gives $A_g > 0$ because $x^0 + g$ is feasible. All minima over empty sets below equal $+\infty$.

Theorem 2 (Continuous and grid thresholds) *Define the thresholds*

$$\tau = \min_{g: D_g > 0} \frac{A_g}{D_g}, \qquad \theta_q = \min_{g: D_g > S_g/q} \frac{A_g}{D_g - S_g/q}. \tag{5}$$

These thresholds govern the optimality of x^0 as follows.

(i) ***Continuous problem.*** *The plan x^0 is the unique minimizer of F_t over P if and only if $0 \le t \le \tau$.*

(ii) ***Grid problem.*** *For each q, the plan x^0 minimizes F_t over P_q if and only if $0 \le t \le \theta_q$. It is the unique minimizer for $0 \le t < \theta_q$. If $\theta_q < \infty$, at least one other grid plan is also optimal at $t = \theta_q$.*

(iii) ***Threshold comparison.*** *If $\tau < \infty$, then $\theta_q > \tau$ for every q, possibly with $\theta_q = \infty$. If $\tau = \infty$, then $\theta_q = \infty$ for every q.*

Proof. At $t = 0$, unique linear optimality gives uniqueness on both P and P_q. Henceforth let $t > 0$. Every feasible direction at x^0 decomposes into a nonnegative combination of cycles in $\mathcal{G}$: orient positive differences forward and negative differences backward, then decompose the resulting circulation. The directional derivative along g is $A_g - tD_g$. Convexity therefore gives continuous optimality exactly when these quantities are nonnegative, equivalently $t \le \tau$: cycles with $D_g \le 0$ impose no restriction because $A_g > 0$. Strict convexity gives uniqueness.

For a grid step, the exact expansion is

$$q\big[F_t(x^0 + g/q) - F_t(x^0)\big] = A_g - tD_g + \frac{t}{q} S_g. \tag{6}$$

Lemma 1, applied with $a_e = tb_e > 0$, gives grid optimality exactly when $t \le \theta_q$. Every cycle inequality is strict when $t < \theta_q$, giving uniqueness. If $\theta_q < \infty$, choose a cycle g attaining its defining minimum. Equation (6) shows that $x^0 + g/q \ne x^0$ is also optimal at $t = \theta_q$.

If $\tau = \infty$, then $D_g \le 0$ for every cycle, so all grid thresholds are infinite. Otherwise, every eligible cycle satisfies

$$\frac{A_g}{D_g - S_g/q} > \frac{A_g}{D_g} \ge \tau.$$

Since $\mathcal{G}$ is finite, the minimum gives $\theta_q > \tau$, trivially if $\theta_q = \infty$. □

The correction tS_g/q in (6) is the cost of a finite step beyond its directional derivative. It delays the first grid transition and can change which cycle determines

that transition. For a fixed t, negative-cycle detection checks the threshold inequalities without listing all simple cycles. The result concerns persistence of x^0; it does not assert monotonicity of each flow along the whole path.

Example 1 (A sharp two-by-two instance) Let $\mu = \nu = (1, 1)$, $b_{ij} = 1$, and $c = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Write $x(s) = \left(\begin{smallmatrix} 1-s & s \\ s & 1-s \end{smallmatrix}\right)$. The unique linear optimum is $x(0)$ and

$$F_t(x(s)) = 2s + t(2 - 4s + 4s^2).$$

The only feasible cycle at $x(0)$ has $(A, D, S) = (2, 4, 4)$. Hence

$$\tau = \frac{1}{2}, \qquad \theta_1 = \infty, \qquad \theta_q = \frac{q}{2(q-1)} \quad (q \geq 2). \tag{7}$$

For $t > 0$, the continuous optimum has $s^{\mathbb{R}}(t) = \max\{0, \frac{1}{2} - \frac{1}{4t}\}$. Grid optima are the nearest feasible grid points to this value, with both choices retained at ties. Figure 1 shows the paths.

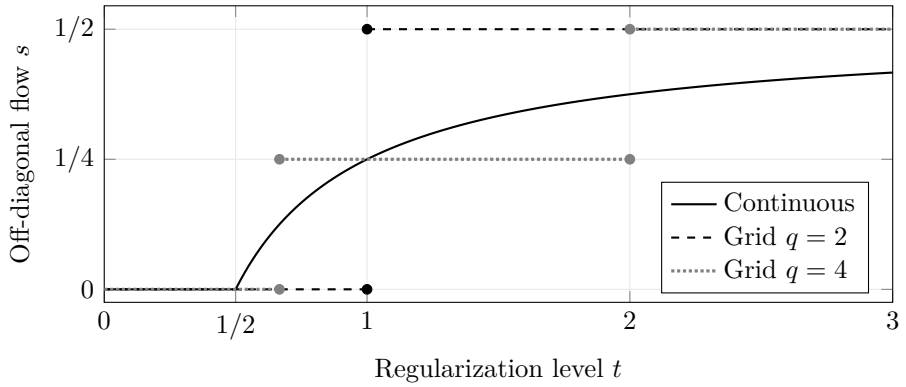


Fig. 1 Off-diagonal flow in Example 1. The continuous path leaves zero at $t = 1/2$, while the grids $q = 4$ and $q = 2$ leave at $2/3$ and 1. Filled endpoints denote ties

3.1 Pure group assignments

Fix $t > 0$ and write $a_e = tb_e$, so $F_t(x) = \sum_e (c_e x_e + a_e x_e^2)$. Suppose $n = m = M$ and ζ is a permutation with $\nu_{\zeta_i} = \mu_i$. Define

$$\bar{x}_{i\zeta_i} = \mu_i, \qquad \bar{x}_{ij} = 0 \quad (j \neq \zeta_i).$$

The next characterization allows unequal group sizes and does not assume that $\bar{x}$ is linear optimal. For distinct groups i, k, set

$$v_{ik} = c_{i\zeta_k} - c_{i\zeta_i} - 2a_{i\zeta_i}\mu_i, \qquad w_{ik}^{(q)} = v_{ik} + \frac{a_{i\zeta_k} + a_{i\zeta_i}}{q}. \tag{8}$$

Theorem 3 (Pure assignments) *Let $q \in \mathbb{Z}_{++}$, and define the group weights v and $w^{(q)}$ by* (8).

(i) ***Grid problem.*** *The plan $\bar{x}$ is optimal over P_q if and only if every directed simple cycle of groups has nonnegative total $w^{(q)}$-weight. It is the unique optimum if and only if every such total is positive.*

(ii) ***Continuous problem.*** *The plan $\bar{x}$ is optimal over P if and only if every directed simple cycle of groups has nonnegative total v-weight. Whenever $\bar{x}$ is optimal over P, it is the unique optimum.*

(iii) ***Sufficient condition.*** *Suppose that, for every group i,*

$$c_{ij} + \frac{a_{ij}}{q} \geq c_{i\zeta_i} + a_{i\zeta_i}\left(2\mu_i - \frac{1}{q}\right), \qquad j \neq \zeta_i. \tag{9}$$

Then $\bar{x}$ is optimal over P_q, uniquely if every inequality is strict.

Proof. At $\bar{x}$, reverse residual arcs exist only on the designated connections (i, ζ_i). A forward arc from group i to destination ζ_k must therefore be followed by the reverse arc to group k. Since ζ is a permutation, this gives a one-to-one correspondence between nontrivial simple residual cycles and directed simple group cycles.

The corresponding move reassigns $1/q$ units of each participating group i from ζ_i to ζ_k. It is feasible since $\mu_i \geq 1/q$, and changes group i's contribution by

$$\frac{c_{i\zeta_k} - c_{i\zeta_i} - 2a_{i\zeta_i}\mu_i}{q} + \frac{a_{i\zeta_k} + a_{i\zeta_i}}{q^2} = \frac{w^{(q)}_{ik}}{q}.$$

Lemma 1 therefore gives the grid optimality and uniqueness criteria. For the continuous problem, the directional derivative along each cycle is its total v-weight. Feasible directions decompose into these cycles, so convexity gives optimality and strict convexity gives uniqueness. Finally, because ζ is a permutation, (9) is equivalent to $w^{(q)}_{ik} \geq 0$ for every $i \neq k$. Summing these inequalities along any group cycle gives a nonnegative total, which is positive when the inequalities are strict. □

Corollary 4 (A sufficient linear-cost gap) *For each i, suppose $a_{ij} \geq a_{i\zeta_i}$ for $j \neq \zeta_i$ and*

$$\min_{j \neq \zeta_i} c_{ij} - c_{i\zeta_i} > a_{i\zeta_i}\mu_i^2(1 - 1/M). \tag{10}$$

Then $\bar{x}$ is the unique optimum over P_1.

Proof. Since $M \geq 2$, $\mu_i^2(1 - 1/M) \geq 2(\mu_i - 1)$, as $\mu_i^2/2 - 2(\mu_i - 1) = (\mu_i - 2)^2/2 \geq 0$. The assumptions therefore imply all strict inequalities in (9) with $q = 1$. □

The coefficient ordering in this sufficient gap condition cannot simply be dropped. With $\mu = \nu = (2, 2)$ and

$$a = \begin{pmatrix} 10 & 1 \\ 1 & 10 \end{pmatrix}, \qquad c = \begin{pmatrix} 1 & 22 \\ 22 & 1 \end{pmatrix},$$

the gap condition holds because $21 > 10 \cdot 4(1 - 1/2) = 20$, but the diagonal plan costs 84 and the all-ones plan costs 68. The exact criterion in Theorem 3 needs no coefficient ordering: here the two-cycle has total $w^{(1)}$-weight -16.

4 Replication and the threshold correction

The normalized grid formulation has a direct population interpretation. Multiplying the objective by q and putting $z = qx$ gives

$$\min_{x \in P_q} F_t(x) = \frac{1}{q} \min_{z \in qP \cap \mathbb{Z}^{n \times m}} \left\{ c \cdot z + \frac{t}{q} Q(z) \right\}. \tag{11}$$

Thus q replicates both margins, with congestion scaled by $1/q$ to keep the objective per replicated population comparable. With an unscaled coefficient on $Q(z)$, the continuous and integer thresholds are τ/q and θ_q/q.

Theorem 5 (Exact replication law) *Suppose $\tau < \infty$. For g with $D_g > 0$, set*

$$\alpha_g = \frac{A_g}{D_g}, \qquad \rho_g = \frac{S_g}{D_g}, \qquad \rho_* = \min_{g:\alpha_g = \tau} \rho_g. \tag{12}$$

Then θ_q is nonincreasing in q, strictly decreasing once finite, and converges to τ. It is finite exactly when $q > \min_{g:D_g>0} \rho_g$. For every integer $q > \rho_*$,

$$0 < \theta_q - \tau \le \frac{\tau \rho_*}{q - \rho_*}. \tag{13}$$

Define the sufficient replication index

$$q_0 = \max \left\{ \lfloor \rho_* \rfloor + 1, \quad \max_{\substack{g:D_g>0 \\ \alpha_g > \tau}} \left\lceil \frac{\alpha_g \rho_* - \tau \rho_g}{\alpha_g - \tau} \right\rceil \right\}, \tag{14}$$

omitting the inner maximum when its index set is empty. For every integer $q \ge q_0$,

$$\theta_q = \frac{\tau}{1 - \rho_*/q}, \qquad \lim_{q \to \infty} q(\theta_q - \tau) = \tau \rho_*. \tag{15}$$

Consequently the unscaled threshold difference satisfies $q^2(\theta_q/q - \tau/q) \to \tau \rho_*$.

Proof. Theorem 2 gives

$$\theta_q = \min_{g:q > \rho_g} \frac{\alpha_g}{1 - \rho_g/q}.$$

Each eligible expression is strictly decreasing in q, and eligibility expands with q. This proves monotonicity and the finiteness criterion. A cycle with $(\alpha_g, \rho_g) = (\tau, \rho_*)$ gives (13), and convergence to τ follows.

Among cycles with $\alpha_g = \tau$, the smallest ρ_g gives the smallest threshold whenever eligible. For an eligible cycle with $\alpha_g > \tau$ and $q > \rho_*$, the comparison

$$\frac{\alpha_g}{1 - \rho_g/q} \ge \frac{\tau}{1 - \rho_*/q} \quad \Longleftrightarrow \quad q \ge \frac{\alpha_g \rho_* - \tau \rho_g}{\alpha_g - \tau}$$

follows by multiplying positive denominators. The choice of q_0 ensures these inequalities and eligibility of a cycle with $(\alpha_g, \rho_g) = (\tau, \rho_*)$; ineligible cycles impose no condition. This proves (15), and expansion gives the limits. □

The reciprocal form makes the changes of minimizing cycle explicit:

$$\frac{1}{\theta_q} = \max \left\{ 0, \max_{g \in \mathcal{G}} \left(\frac{D_g}{A_g} - \frac{S_g}{q A_g} \right) \right\}, \qquad \frac{1}{\infty} = 0. \tag{16}$$

It is the restriction of a convex piecewise-affine function of the grid size $1/q$. The monotonicity in Theorem 5 therefore holds even though successive grid sets need not be nested. In Example 1, $\rho_* = 1$ and $\theta_q - \tau = 1/[2(q-1)]$ for $q \geq 2$. Thus (13) holds with equality, and the first-order rate cannot be improved.

Example 2 (The critical cycle can change) Take $\mu = (2, 2)$, $\nu = (1, 1, 2)$, and

$$b = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \end{pmatrix}, \qquad c = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 1 & 0 \end{pmatrix}, \qquad x^0 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

The unique linear optimum x^0 has two feasible simple cycles. Their (A, D, S) values are $(2, 10, 6)$ and $(1, 6, 6)$, respectively. Thus $\tau = 1/6$ and $\rho_* = 1$, while

$$\theta_q = \begin{cases} 2q/(10q-6), & q = 1, 2, \\ q/[6(q-1)], & q \geq 3. \end{cases} \tag{17}$$

Both cycles attain the threshold at $q = 3$, which is also the sufficient index q_0 in (14). The cycle determining the continuous threshold therefore need not determine the integer threshold on a coarse grid.

Remark 1 (Bounded network flows) The cycle argument also applies to a nonempty bounded flow polytope $\{x : Ax = d,\ \ell \leq x \leq u\}$ with network incidence matrix A and integer data. Use feasible signed residual cycles and a unique linear optimum. Replication must scale balances and both bounds; keeping capacities fixed would invalidate (11).

5 Monotone comparative statics

Monotonicity holds for aggregate congestion and for a connection's response to its own cost. It need not hold for each flow along the common regularization path. The statements below apply to every optimal selection and do not assume that the integer plan is unique.

Proposition 6 (Monotone responses) *For $F(x) = c \cdot x + \sum_e a_e x_e^2$, $a_e > 0$, on either P or a fixed grid P_q, increasing only c_e or only a_e weakly decreases the optimal flow x_e. For $0 \leq t_1 < t_2$ and any minimizers x^1, x^2 of F_{t_1}, F_{t_2} over the same feasible set,*

$$Q(x^2) \leq Q(x^1), \qquad c \cdot x^2 \geq c \cdot x^1. \tag{18}$$

Proof. If x^r minimizes $G(x) + s_r T(x)$ and $s_1 < s_2$, adding the two optimality inequalities gives $(s_2 - s_1)[T(x^2) - T(x^1)] \leq 0$. Take $T(x) = x_e$, x_e^2, or $Q(x)$; nonnegative flows give the own-coefficient claims. Optimality at t_1 also gives $c \cdot (x^2 - x^1) \geq t_1[Q(x^1) - Q(x^2)] \geq 0$. □

These order comparisons follow directly from optimality. Integer flow structure gives the more precise unit adjustments below. Fix positive coefficients a_e on P_1, and vary only the linear cost of connection e, writing $c_e = s$. Define

$$H_e(k) = \min_{x \in P_1 : x_e = k} \left\{ a_e x_e^2 + \sum_{f \neq e} (c_f x_f + a_f x_f^2) \right\}. \tag{19}$$

For $e = (i, j)$, the feasible counts run from $L_e = \max\{0, \mu_i + \nu_j - \sum_h \mu_h\}$ to $U_e = \min\{\mu_i, \nu_j\}$. These bounds are attainable, and every intermediate integer count extends to a feasible table by completing its row and column and then the remaining margins.

Proposition 7 (Unit adjustments) *For $L_e < k < U_e$,*

$$H_e(k-1) + H_e(k+1) - 2H_e(k) \geq 2a_e. \tag{20}$$

Set $\sigma_k = H_e(k-1) - H_e(k)$ for $L_e < k \leq U_e$, with $\sigma_{L_e} = +\infty$ and $\sigma_{U_e+1} = -\infty$. The optimal count is k exactly when $\sigma_{k+1} \leq s \leq \sigma_k$. At a finite threshold σ_k, precisely $k-1$ and k are optimal counts, and consecutive finite thresholds are separated by at least $2a_e$. Thus, for $s_1 < s_2$ and any associated optimal counts k_1, k_2,

$$0 \leq k_1 - k_2 \leq 1 + \frac{s_2 - s_1}{2a_e}. \tag{21}$$

Proof. Let x, y attain $H_e(k-1), H_e(k+1)$ and denote the objective in (19) by Φ. A cycle decomposition of $y - x$ contains a signed simple cycle g with $g_e = 1$ conforming to its signs. Both $x + g$ and $y - g$ are feasible and have coordinate e equal to k. Quadratic expansion gives

$$\Phi(x) + \Phi(y) - \Phi(x+g) - \Phi(y-g) = 2 \sum_{f : g_f \neq 0} a_f(|y_f - x_f| - 1) \geq 2a_e.$$

Since $2H_e(k) \leq \Phi(x+g) + \Phi(y-g)$, this proves (20). The differences of $H_e(k) + sk$ are strictly increasing, so the stated neighboring inequalities characterize its minimum. A decline by $d \geq 1$ crosses d thresholds separated by at least $2a_e$; hence $s_2 - s_1 \geq 2a_e(d-1)$. □

This exchange argument is a quadratic specialization of the discrete convexity of flow value functions (Murota, 2016; Murota & Shioura, 2005). The chosen count is unique away from its thresholds; the whole plan may still have ties.

Increasing only $a_e > 0$ also produces downward unit steps. Indeed, the optimized cost at each fixed count k is affine in a_e, with slope k^2, so a change of optimal count occurs at a tie. Applying (20) at that positive parameter shows that only adjacent counts can tie. The direction of each step follows from Proposition 6.

Example 3 (A nonmonotone flow along the regularization path) Let $\mu = (2, 2)$, $\nu = (1, 1, 2)$, and

$$b = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix}, \qquad c = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The four integer plans have first rows $(0, 0, 2)$, $(1, 0, 1)$, $(0, 1, 1)$, and $(1, 1, 0)$; their costs are, respectively, $11t$, $1 + 6t$, $2 + 5t$, and $3 + 6t$. The first three are uniquely optimal on $[0, 1/5)$, $(1/5, 1)$, and $(1, \infty)$, with neighboring plans tied at the endpoints. Thus x_{11} follows $0, 1, 0$, while Q falls from 11 to 6 to 5.

6 Approximation along the regularization path

Write $V_q(t) = \min_{P_q} F_t$ and $V_{\mathbb{R}}(t) = \min_P F_t$. The next bound is a quadratic consequence of feasible dependent rounding (Gandhi et al., 2006). It complements the first-transition formulas by controlling the entire path.

Proposition 8 (Value and plan bounds) *Fix $t > 0$, let $x^{\mathbb{R}} = x^{\mathbb{R}}(t)$, and put $r_e = qx_e^{\mathbb{R}} - \lfloor qx_e^{\mathbb{R}} \rfloor$. The optimal values then satisfy*

$$0 \le V_q(t) - V_{\mathbb{R}}(t) \le \frac{t}{q^2} \sum_e b_e r_e (1 - r_e) \le \frac{t}{4q^2} \sum_e b_e. \tag{22}$$

For every grid optimum x^q,

$$\sum_e b_e (x_e^q - x_e^{\mathbb{R}})^2 \le \frac{V_q(t) - V_{\mathbb{R}}(t)}{t} \le \frac{1}{4q^2} \sum_e b_e. \tag{23}$$

The value bound is attained or improved by a feasible rounding satisfying $\lfloor qx^{\mathbb{R}} \rfloor \le qx \le \lceil qx^{\mathbb{R}} \rceil$. Its least-cost choice is a binary transportation problem.

Proof. The polytope

$$\{z \in qP : \lfloor qx^{\mathbb{R}} \rfloor \le z \le \lceil qx^{\mathbb{R}} \rceil\}$$

contains $qx^{\mathbb{R}}$ and has integer vertices by total unimodularity. Express $qx^{\mathbb{R}}$ as a convex combination of these vertices, and let Z have the resulting distribution. Then $Y = Z/q$ satisfies $\mathbb{E}Y = x^{\mathbb{R}}$ and $\mathrm{Var}(Y_e) = r_e(1 - r_e)/q^2$. Therefore

$$\mathbb{E}F_t(Y) - F_t(x^{\mathbb{R}}) = \frac{t}{q^2} \sum_e b_e r_e (1 - r_e).$$

Some vertex costs no more than this average. Writing $Z = \lfloor qx^{\mathbb{R}} \rfloor + w$ leaves binary choices and integer residual margins; the cost increments are linear in w. Finally, continuous first-order optimality and quadratic expansion give

$$F_t(x^q) - F_t(x^{\mathbb{R}}) = \nabla F_t(x^{\mathbb{R}}) \cdot (x^q - x^{\mathbb{R}}) + t \sum_e b_e (x_e^q - x_e^{\mathbb{R}})^2 \ge t \sum_e b_e (x_e^q - x_e^{\mathbb{R}})^2.$$

This proves (23). □

The distance bound is uniform over $t > 0$ and over all grid optima. It does not imply that the best rounding is a global grid optimum. In Example 1 with $t = 1$, the continuous optimum is $x^{\mathbb{R}} = \left(\begin{smallmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{smallmatrix}\right)$. For every $q \equiv 2 \pmod 4$, nearest grid points have $|s - s^{\mathbb{R}}| = 1/(2q)$, and therefore

$$V_q(1) - V_{\mathbb{R}}(1) = \frac{1}{q^2}, \qquad \sum_e b_e (x_e^q - x_e^{\mathbb{R}})^2 = \frac{1}{q^2}. \tag{24}$$

Since $\sum_e b_e = 4$, both constants in (22)–(23) are sharp. In the same instance, $\theta_q - \tau$ is of order $1/q$. Hence a value error of order $1/q^2$ can coexist with an error of order $1/q$ in the regularization level at which the allocation changes.

There is also a simple condition for exact integrality. If the positive support of $x^{\mathbb{R}}(t)$ is a forest, a leaf's integer margin determines its only incident flow as an integer; successive leaf removal proves $x^{\mathbb{R}}(t) \in P_1$. For fixed $t > 0$ and b, a continuous integer optimum z with a positive-support cycle g instead requires

$$\sum_e g_e (c_e + 2tb_e z_e) = 0.$$

These equalities form a finite union of proper hyperplanes in c. Hence continuous optima that are integral and have cyclic positive support are exceptional with respect to linear costs.

A Additional structural results

A.1 Existence and the number of feasible plans

Proposition A.1 (Existence and finite grids) *For balanced positive integer margins with total N, the polytope P is nonempty and compact, and each P_q is nonempty and finite. In particular,*

$$|P_q| \le \prod_{i=1}^{n} \binom{q\mu_i + m - 1}{m-1} \le m^{qN}. \tag{25}$$

Every continuous objective attains its minimum on P, and every real-valued objective attains its minimum on P_q. A strictly convex objective has a unique minimizer on P, and a linear objective has an integral one.

Proof. The matrix $\mu\nu^T/N$ proves continuous feasibility; successively matching available integer supply and demand constructs a plan in P_1. The constraints are closed and $0 \le x_{ij} \le \min\{\mu_i, \nu_j\}$, proving compactness. Counting nonnegative compositions of each row of qx while ignoring column restrictions gives the product in (25). For an integer $r \ge 1$, $\binom{r+m-1}{m-1} = \prod_{h=1}^{r}(h+m-1)/h \le m^r$. The remaining claims follow from compactness, finiteness, strict convexity, and total unimodularity, respectively. □

A.2 Continuous solution and smooth sensitivity

Consider $F(x) = c \cdot x + \sum_e a_e x_e^2$, $a_e > 0$; for the main model, $a_e = tb_e$ at fixed $t > 0$. Vectorize x, set $H = 2\,\mathrm{diag}(a)$, and let $\widetilde{R} \in \mathbb{R}^{(n+m)\times nm}$ contain all row and column equations. Let R omit the last column equation and put $d = (\mu_1, \dots, \mu_n, \nu_1, \dots, \nu_{m-1})^T$.

Proposition A.2 (Redundancy and the reduced system) *Writing $v = (\mathbf{1}_n, -\mathbf{1}_m)^T$, the full systems satisfy*

$$\ker(\widetilde{R}H^{-1}\widetilde{R}^T) = \mathrm{span}\{v\}, \qquad \mathrm{rank}(\widetilde{R}H^{-1}\widetilde{R}^T) = n + m - 1, \tag{26}$$

$$\ker \begin{pmatrix} H & -\widetilde{R}^T \\ -\widetilde{R} & 0 \end{pmatrix} = \mathrm{span}\{(0, v)\}. \tag{27}$$

The matrix $S = RH^{-1}R^T$ is positive definite, and the reduced KKT matrix is nonsingular. The equality-constrained minimizer is

$$\widehat{x} = H^{-1}\left[R^T S^{-1}(d + RH^{-1}c) - c\right]. \tag{28}$$

If $\widehat{x} \ge 0$, it is the unique minimizer over P.

Proof. The equations $\widetilde{R}^T(u, w) = 0$ are $u_i + w_j = 0$ for every connection, so $\ker \widetilde{R}^T = \mathrm{span}\{v\}$. Positive definiteness of H gives (26). If (z, p) belongs to the full KKT nullspace, then $Hz = \widetilde{R}^T p$ and $\widetilde{R}z = 0$; multiplying by z^T gives $z = 0$, then $p \in \mathrm{span}\{v\}$. Deleting one redundant equation leaves R with full row rank, so $S \succ 0$ and the reduced KKT matrix is nonsingular. Solving $Hx + c = R^T p$, $Rx = d$ yields (28). If this vector is nonnegative, it also minimizes over P. □

Thus singularity of the full system reflects only the multiplier normalization $(u, w) \mapsto (u + \alpha\mathbf{1}_n, w - \alpha\mathbf{1}_m)$. It does not obstruct an explicit solution or local comparative statics. If $\widehat{x}$ has negative entries, the active nonnegativity constraints must be included; coordinatewise clipping need not preserve the margins.

Corollary A.3 (Local derivatives and the envelope) *On a region where the continuous optimum is strictly positive, define*

$$K = H^{-1} - H^{-1}R^T S^{-1} R H^{-1} \succeq 0. \tag{29}$$

With $\mathbf{e}_e$ the coordinate vector for connection e,

$$\frac{\partial x}{\partial c} = -K, \qquad \frac{\partial x}{\partial a_e} = -2x_e K \mathbf{e}_e, \qquad \frac{\partial x}{\partial d} = H^{-1}R^T S^{-1}. \tag{30}$$

In particular, $\partial x_e/\partial c_e \leq 0$ and $\partial x_e/\partial a_e \leq 0$. For $V(a,c) = \min_{x \in P} F(x)$, including at boundary optima,

$$\frac{\partial V}{\partial c_e} = x_e, \qquad \frac{\partial V}{\partial a_e} = x_e^2. \tag{31}$$

Proof. Differentiating the reduced KKT equations gives (30). Also $K = H^{-1/2}(I - \mathcal{P})H^{-1/2}$, where $\mathcal{P}$ is the orthogonal projector onto the row space of $RH^{-1/2}$. Thus $K_{ee} \geq 0$, giving the own-cost signs. The envelope identities follow from unique minimization on the fixed compact set P; they do not require an interior optimum. □

These are the reduced equality-constrained quadratic formulas (Boyd & Vandenberghe, 2004, Chapters 5 and 10). On a fixed active face, the same calculation uses its free variables and independent constraints. Margin perturbations must preserve feasibility on that face. Support changes can create kinks in the optimizer. In the main parametrization, the envelope gives $\partial V_{\mathbb{R}}/\partial b_e = t(x_e^{\mathbb{R}})^2$ and $dV_{\mathbb{R}}/dt = Q(x^{\mathbb{R}})$ for $t > 0$.

A.3 The general two-by-two solution

Proposition A.4 (Closed-form continuous and grid solutions) *For $n = m = 2$, every feasible plan has the form*

$$x(s) = \begin{pmatrix} s & \mu_1 - s \\ \nu_1 - s & \mu_2 - \nu_1 + s \end{pmatrix}, \qquad L \leq s \leq U,$$

where $L = \max\{0, \nu_1 - \mu_2\}$ and $U = \min\{\mu_1, \nu_1\}$. Put

$$\begin{aligned} \kappa &= a_{11} + a_{12} + a_{21} + a_{22}, \\ \ell &= c_{11} - c_{12} - c_{21} + c_{22} - 2a_{12}\mu_1 - 2a_{21}\nu_1 + 2a_{22}(\mu_2 - \nu_1). \end{aligned} \tag{32}$$

Then the continuous minimizer is $x(s^{\mathbb{R}})$, where

$$s^{\mathbb{R}} = \min\{U, \max\{L, -\ell/(2\kappa)\}\}. \tag{33}$$

The optimizers over P_q are the plans whose s is nearest to $s^{\mathbb{R}}$ in $q^{-1}\mathbb{Z} \cap [L, U]$, with both choices included at a tie.

Proof. Substitution gives $F(x(s)) = \kappa s^2 + \ell s + \text{constant}$. Since $\kappa > 0$, its vertex is $-\ell/(2\kappa)$; clipping gives the continuous minimizer. The endpoints L, U belong to every grid because the margins are integer. Completing the square therefore gives the stated grid rule, including when the vertex lies outside $[L, U]$. □

The formula covers both interior and boundary allocations. With $a_{ij} = 2$, $\mu = (20, 20)$, $\nu = (12, 28)$, and $c = \left(\begin{smallmatrix} 24 & 48 \\ 16 & 24 \end{smallmatrix}\right)$, it gives $\kappa = 8$, $\ell = -112$, and the common continuous and integer optimum $\left(\begin{smallmatrix} 7 & 13 \\ 5 & 15 \end{smallmatrix}\right)$. With $a = c = \left(\begin{smallmatrix} 200 & 2 \\ 2 & 200 \end{smallmatrix}\right)$ and $\mu = \nu = (10, 10)$, the unconstrained vertex is $-79/202$, so the common optimum is $\left(\begin{smallmatrix} 0 & 10 \\ 10 & 0 \end{smallmatrix}\right)$. Online Resource 1 also verifies a four-by-four pure assignment using Theorem 3.

Statements and Declarations

Data and code availability. All examples use synthetic data specified in the article, and no external dataset is required. Exact verification scripts and their outputs are provided in Online Resource 1.

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